[INCH-POUND]
A-A-59336
28 September 1999
SUPERSEDING
MIL-S-24034E
29 April 1988
MIL-S-80201C
6 July 1987

COMMERCIAL ITEM DESCRIPTION

SAW, BAND, METAL CUTTING, VERTICAL, CONTOUR

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-S-24034E and MIL-S-80201C for all federal agencies.

1. SCOPE. This commercial item description (CID) covers vertical contour metal-cutting bandsaws with both power-feeding and nonpower-feeding worktables. Vertical contour bandsaws are used for metal contour cutting operations and internal cutting operations whereby the saw blade is cut, inserted into the workpiece pilot hole, and then welded back together for the sawing operation.

2. CLASSIFICATION. The bandsaws shall be of the following types and sizes. Size refers to the maximum distance from the cutting point of the saw blade to the bandsaw column (throat depth), the maximum distance from the table to the blade guides (throat height), and powered material feed travel (stroke). The type and size to be furnished shall be as specified (see 7.3(b)).

Type I - Manual material feed
Type II - Powered material feed

Size - (see table I)

3. SALIENT CHARACTERISTICS

3.1 General requirements. The bandsaw shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3405

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TABLE I. Vertical bandsaw sizes.

Size	Type I		Type II		
	Throat depth, inches	Throat height, inches	Throat depth, inches	Throat height, inches	Stroke, inches
14	14	11	-	-	-
20	20	12	20	10	8.5
26	26	13	26	12	8.5
36	36	13	36	13	12

Note 1: Size 14 is not available in type II band saws.
Note 2: Band saw sizes and capacities shall be not less than the stated requirements of the size ordered shown in above table. When the largest size and capacity band saw shown in above table is ordered, the size and capacity of the band saw offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

3.2 Components. The bandsaw shall include, as a minimum, the following components:

3.2.1 Base and column. The base and column shall contain all the components of the contour vertical bandsaw. Access doors or covers shall be provided for blade changes, adjustments, and servicing. The base shall have a means to level and anchor the bandsaw. The throat depth shall correspond to the applicable size shown in table I.

3.2.2 Worktable. Unless otherwise specified (see 7.3(c)), a tilting worktable that can be pivoted and securely locked at any angle ranging from 45 degrees right to 5 degrees left shall be provided. When specified (see 7.3(d)), an auxiliary worktable shall be provided for the type I bandsaw. Unless otherwise specified (see 7.3(e)), the type II bandsaw shall have hydraulically powered material feed capability. The stroke of the powered material feed shall correspond to the applicable size shown in table I. If required, the type II bandsaw shall be provided with contour and circular cutting attachments as specified (see 7.3(f)).

3.2.3 Drive and idle wheels. A variable-speed blade drive and idler wheel with a means for blade tensioning shall be provided. When specified (see 7.3(g)), blade speed and tension indicators shall be provided. Unless otherwise specified (see 7.3(h)), the blade drive motor shall be electric.

3.2.4 Band welder and annealer. Unless otherwise specified (see 7.3(i)), the bandsaw shall be equipped with a band welder and annealer, band clamp, shears, and grinder. Unless otherwise specified (see 7.3(j)), the band welder and annealer shall be mounted in the column of the bandsaw and shall accommodate the maximum blade width for the bandsaw specified.

3.2.5 Guidepost. A blade guidepost shall be provided and shall contain the upper blade guides. The guidepost shall be vertically adjustable to accommodate the maximum workpiece thickness required. The throat height shall correspond to the applicable size shown in table I. The blade guides shall consist of carbide-faced roller bearings and shall be constructed with the strength and rigidity to meet the cutting requirements shown in tables II and III.

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3.2.6 Saw blades and attachments. Saw blades of the specified material, quantity, size, tooth pitch, and type shall be furnished with each bandsaw as specified (see 7.3(k)). When specified (see 7.3(l)), the bandsaw shall be provided with band filing and polishing attachments and file bands.

3.2.7 Controls. The bandsaw controls shall be convenient to the operator. Unless otherwise specified (see 7.3(m)), controls shall consist of start/stop, blade speed, and blade tension. The type II bandsaw shall be provided with power feed controls and attachments as specified (see 7.3(n)) to assist the operator when manipulating workpieces for contour cuts.

3.2.8 Coolant system. Unless otherwise specified (see 7.3(o)), the type II bandsaw shall be provided with a coolant system. Means shall be provided to determine and maintain coolant level within the sump. A removable strainer shall be provided to remove metal chips from the coolant.

3.2.9 Hydraulic system. When furnished, the hydraulic system shall be provided with a fluid level indicator and overpressure protection.

3.2.10 Electrical system. The electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79, "Machinery, Industrial, Electrical Standard For". Unless otherwise specified (see 7.3(p)), the bandsaw shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit; and shall be initially wired for 230-volt operation. The control system shall operate on a 120-volt, single-phase, 60-hertz circuit. The electrical circuitry shall provide low voltage and overload protection.

3.2.11 Electrical motors. The vertical bandsaw electric drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.3(q)), the motor(s) shall have open dripproof enclosures. The motor(s) shall have a 30-minute duty cycle horsepower rating.

3.2.12 Optional accessories. Optional accessories shall be furnished as specified (see 7.3(r)).

3.3 Performance. Unless otherwise specified (see 7.3(s)), the bandsaw shall be capable of producing cuts in SAE 1018 steel that comply with the requirements shown in tables II and III when tested in accordance with sections 5.3.1 through 5.3.4.

TABLE II. Type I bandsaw performance requirements.

Size	Cutting rate, square inches per minute	Perpendicularity, inch per inch of thickness
14	8	0.008
20	8	0.005
26	12	0.005
36	12	0.005

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TABLE III. Type II bandsaw performance requirements.

Size	Perpendicularity, inch per inch of thickness	Cutting rate, square inches per minute	
		Straight cut	Contour cut
20	0.002	8	5
26	0.002	12	5
36	0.002	12	6

3.4 Bandsaw dimensions and weight. If required, the maximum bandsaw dimensions (length, width, and height) and bandsaw weight shall not exceed the restrictions specified (see 7.3(t)).

3.5 Safety and health requirements. The bandsaw shall be designed and manufactured in accordance with ANSI B11.10, "Sawing Machines, Metal, Safety Requirements for the Construction, Care and Use Of". The manufacturer shall ensure the bandsaw and all equipment and accessories used on the bandsaw shall comply with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the OSHA and ANSI standards, the OSHA standards shall apply.

3.6 Systems of units. The US Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.3(u)). Regardless of the measurement system used, all measuring and indicating devices on the bandsaw shall be graduated in the same system. When specified (see 7.3(v)), measuring and indicating devices shall be graduated in both the US and SI systems of measurement.

3.7 Nameplate. A nameplate shall be securely attached to the bandsaw. Unless otherwise specified (see 7.3(w)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Serial number
d. Bandsaw model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.3(x)), a lubrication plate or chart shall be attached to the bandsaw. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products offered shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The contour vertical bandsaw shall be inspected for compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.3(y)), the contour vertical bandsaw shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.3 (z)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.3(aa)).

5.3.2 Operational test. The contour vertical bandsaw shall be operated in accordance with the manufacturer's standard operating procedures. The contour vertical bandsaw shall be operated at no load for not less than 15 minutes. Proper operation of all controls and features shall be verified during the trial period. For the tests described herein, the choice of saw blade, coolant, blade speed, and cutting force shall be the option of the supplier.

5.3.3 Performance tests. The type I bandsaw shall undergo a straight cut test. The worktable shall be clamped in the zero tilt position. The test workpiece shall be SAE 1018 steel and shall have a 1- by 4-inch cross section. The cutting speed and perpendicularity of the cut shall be in accordance with table I. The type II bandsaw shall undergo straight and contour cut tests. The workpiece for the straight cut test shall be an SAE 1018 steel bar and shall have a 4- by 9-inch cross section. For the contour cut test, a 6-inch radius semicircle shall be scribed on the surface of a 2-inch thick SAE 1018 steel plate. A cut shall be made along the scribe and shall not deviate from the scribe by more than 3/16 inch.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.3(bb)).

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5.4 Acceptance. Unless otherwise specified (see 7.3(cc)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The test results will be compared to the performance requirements shown in tables II and III. Failure of the bandsaw to meet these requirements shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. Unless otherwise specified (see 7.3(dd)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

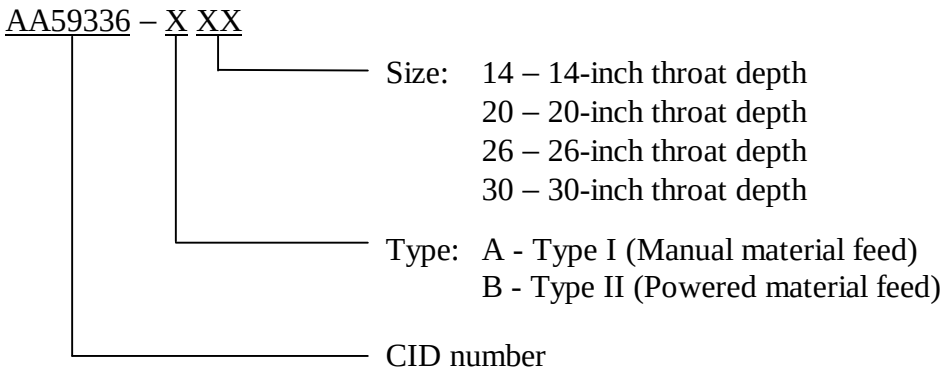
7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



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7.3 Ordering data. Acquisition documents must specify the following information:

a. Title, number and date of this document
b. Type and size of contour vertical bandsaw required (see 2)
c. Tilting worktable, if different (see 3.2.2)
d. Auxiliary worktable, if required (see 3.2.2)
e. Shop hydraulically powered material feed, if different (see 3.2.2)
f. Contour and circular cutting attachments, if required (see 3.2.2)
g. Blade speed and tension indicator, if required (see 3.2.3)
h. Drive motor, if different (3.2.3)
i. Band welder and annealer, if different (see 3.2.4)
j. Location of band welder and annealer, if different (see 3.2.4)
k. Saw blade quantity, size, tooth pitch, and type, as specified (see 3.2.6)
l. Band filing and polishing attachments and file bands, if required (see 3.2.6)
m. Control system, if different (see 3.2.7)
n. Power feed controls or attachments, if required (see 3.2.7)
o. Type II bandsaw coolant system, if different (see 3.2.8)
p. Electrical system voltage requirements, if different (see 3.2.10)
q. Motor enclosure, if different (see 3.2.11)
r. Optional accessories, as specified (see 3.2.12)
s. Performance, if different (see 3.3)
t. Bandsaw dimensions and weight, as specified (see 3.4)
u. System of units required (see 3.6)
v. Dual system of units, if required (see 3.6)
w. Nameplate, if different (see 3.7)
x. Lubrication plate or chart, if required (see 3.8)
y. Testing, if different (see 5.3)
z. Test responsibility, if different (see 5.3.1)
aa. Certification of calibration, as specified (see 5.3.1)
bb. Optional tests, as specified (see 5.3.4)
cc. Acceptance, if different (see 5.4)
dd. Packaging requirements, if different (see 6)

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - CR4
Navy - SH

Reviewers:
Air Force - 84
Navy - MC

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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